

Data Collection Report

Mobile Money Transaction Analysis — CSC 3221 Final Project

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1. Student Information

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2. Sampling Strategy

Data collection was conducted using a convenience sampling approach, targeting a diverse population of mobile money users across Cameroon. Each group member recruited a minimum of 10 voluntary participants from their personal and professional networks, aiming for variety in age, occupation, income range, and geographic zone. Participation was entirely voluntary, preceded by explicit informed consent. A bilingual Google Form (English/French) served as the primary data collection instrument. Participants were guided to export MTN Mobile Money and/or Orange Money transaction SMS history using third-party applications — SMS Exporter on Android, or Text Message Export on iOS — and upload the resulting CSV/Excel file directly through the form. Demographic and contextual information was collected via structured form fields within the same submission flow.

Sample Snapshot

Total responses collected	12 participants
Gender	60% Male, 40% Female
Dominant age group	18-25 (60%), 26-35 (25%), 46-60 (15%)
Occupations	Students (50%), Salaried employees (33%), Self-employed (17%)

Education level	100% university-educated (50% undergraduate, 50% postgraduate)
Geographic zone	Urban - city centre (75%), Suburban (25%)
Mobile money platforms	Both MTN MoMo & Orange Money (42%), Orange Money only (42%), MTN MoMo only (16%)
Primary Mobile money usage	Both personal & business (60%), Personal only (30%), Business only (10%)
Smartphone ownership	100% of respondents

3. Data Trends Observed

Despite the early-stage sample size, several preliminary patterns are already visible from the form responses:

- Multi-platform behaviour is the norm: half of respondents use both MTN MoMo and Orange Money, suggesting that dual-wallet usage is common — likely driven by network-specific merchant and peer ecosystems.
- Mobile money serves mixed purposes: 60% of users report combining personal and business transactions through the same mobile money account, reflecting the blurred line between informal commerce and personal finance in the Cameroonian context.
- The sample is young and urban: 80% of respondents are city-centre based and 50% are students aged 18-25, consistent with the peer-network recruitment method. This represents a limitation in geographic and socioeconomic diversity that will be addressed in further collection rounds.
- Income data is the most sensitive field: 30% of respondents declined to provide their income range — the highest non-response rate across all questions — confirmation that financial disclosure carries social sensitivity even under anonymised conditions.
- Household composition is polarized: 40% of respondents live alone and 40% live in households of 7 or more, with limited representation in the 2-6 range. This likely reflects the student versus professional split in the current sample.

4. Challenges Encountered

- Export process complexity: Neither Android nor iOS native messaging apps support SMS export out of the box. Guiding participants through third-party app installation required significant individual assistance, particularly for users less comfortable with technical steps.
- iOS accessibility barrier: The MsgKeep application for iOS is a paid app, which effectively excluded iPhone users from participating. This introduced an unintended platform bias — all successful CSV/Excel submissions came from Android users.
- Trust and data sensitivity: Despite clear consent procedures and explicit anonymization guarantees, several potential participants were initially reluctant to share their transaction SMS history. One-on-one reassurance about the academic context and the absence of commercial use was often required.
- Technical export limitation: The SMS Exporter app would not allow local file storage after export, only offering in-app storage. Some individuals were assisted on how to get the files out of the app and complete their submission.

5. Ethics & Privacy

- Informed consent: Every participant explicitly agreed to a consent statement before any data was collected, covering the study purpose, what was being collected, storage and use policies, and the right to withdraw.
- Anonymization: All submitted CSV/Excel files will undergo a systematic anonymization pass during the data cleaning phase. Phone numbers, names and any other personal identifiers will be replaced with pseudonymous IDs (e.g. USER_001). No raw files will be published or shared externally.
- Data minimization: Only the transaction SMS export was requested — no account credentials, PINs, or full account numbers. The SMS format captures transaction metadata only.
- Secure storage: All collected files are stored in a restricted Google Drive folder accessible solely to project team members. The form was configured to not collect respondent email addresses.
- No commercial use: Participants were explicitly informed that their data would be used exclusively for this academic project.

6. Tools Used

Data collection form	Google Forms (bilingual En/Fr)
Response & file storage	Google Sheets + Google Drive (restricted access)
SMS export - Android	SMS Exporter by Smart Positive (Google Play Store)
SMS export - iOS	Text Message Export - MsgKeep (App Store, paid — limited uptake)
Data format	CSV (comma-separated values), xlsx (excel)
Anonymization (planned)	Python / Pandas — to be applied in cleaning phase

7. Quality Assurance

Each submitted CSV/Excel file was reviewed upon receipt to confirm it contained valid mobile money SMS entries in the expected MTN MoMo or Orange Money format. Submissions flagged as potentially malformed or unreleased were followed up with the contributor directly.

A two-part confirmation checklist at the end of the form required participants to actively confirm:

1. That the uploaded file contained only their own mobile money SMS.
2. That they consented to anonymized academic use of the data.